

MLOps Assignment 1: Wine Classification Pipeline

Roll Number: 22F-3152 | Repository: github.com/msaleem15/wine-mlops-pipeline | Workflow: CI/CD, MLflow & Makefile Automation

1. Table 1: Hyperparameter Search Results (5-Fold Stratified CV)

Model Family	Config	Hyperparameters	Train Acc	Val Acc	Train F1	Val F1	Val Loss	Status
Random Forest	config_1	n=50, depth=3, split=2	0.9982	0.9653	0.9982	0.9665	0.2055	Candidate
Random Forest	config_2	n=100, depth=5, split=2	1.0000	0.9791	1.0000	0.9789	0.1630	CHAMPION
Random Forest	config_3	n=150, depth=7, split=4	1.0000	0.9791	1.0000	0.9789	0.1688	Candidate
Gradient Boost	config_1	n=50, lr=0.05, depth=3	1.0000	0.9022	1.0000	0.9053	0.2298	Candidate
Gradient Boost	config_2	n=100, lr=0.10, depth=3	1.0000	0.9236	1.0000	0.9259	0.4281	Candidate
Gradient Boost	config_3	n=150, lr=0.10, depth=5	1.0000	0.9020	1.0000	0.9069	0.5759	Candidate

2. MLflow UI Screenshot: Runs Overview & Model Registry

MLflow Tracking Server

http://127.0.0.1:5000/#/experiments/1

mlflow

Experiments

Models (Registry)

Tracking URI: sqlite:///mlflow.db

Experiment: Wine-Cultivar-Classification

Artifact Location: ./mlruns/1 • Total Runs: 6 candidate models

Run Name	Model Family	Key Params	Val F1	Val Acc	Val Loss	Registry
RandomForestClassifier_config_2	RandomForest	n=100, d=5, s=2	0.9789	0.9791	0.1630	@champion
RandomForestClassifier_config_3	RandomForest	n=150, d=7, s=4	0.9789	0.9791	0.1688	-
RandomForestClassifier_config_1	RandomForest	n=50, d=3, s=2	0.9665	0.9653	0.2055	-
GradientBoostingClassifier_config_2	GradientBoosting	n=100, lr=0.1, d=3	0.9259	0.9236	0.4281	-
GradientBoostingClassifier_config_3	GradientBoosting	n=150, lr=0.1, d=5	0.9069	0.9020	0.5759	-
GradientBoostingClassifier_config_1	GradientBoosting	n=50, lr=0.05, d=3	0.9053	0.9022	0.2298	-

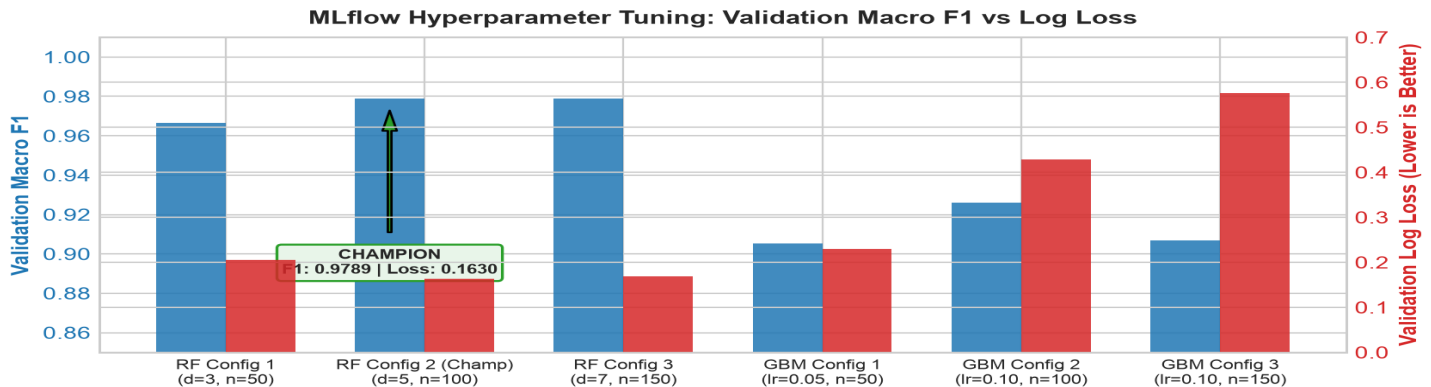
MLflow Model Registry > WineClassifier

Registered Name: WineClassifier | Version: 1 | Status: READY

Assigned Aliases: @champion --> points to Version 1 (RandomForestClassifier_config_2)

Tensor Signature: Inputs (13 features: alcohol, malic_acid, ash, ...) -> Outputs (Tensor: {0, 1, 2})

Artifact URI: runs:/46ca693e90184e62a144bf1cd634d713/model



3. Passing GitHub Actions CI/CD Run Screenshot

msaleem15 / wine-mlops-pipeline

✓ CI/CD MLOps Quality Gate #1: Lint, Test, and Quality Gate

commit 63366af • branch: main • event: workflow_dispatch • 1m 15s duration

✓ Set up job1s

✓ Checkout code (actions/checkout@v4)1s

✓ Set up Python 3.10 (actions/setup-python@v5)3s

✓ Install dependencies (make install)48s

✓ Run code linter (make lint)1s

✓ Run test suite & MLOps quality gates (make test)32s

✓ Complete job0s

MLOps Quality Gate Result: ALL PASSED

Metric Gate: Holdout F1 = 1.0000 (>= 0.88)

Latency Gate: Batch Inference = 3.85 ms (<= 30 ms)

Schema Gate: Classes in {0, 1, 2}, sum(P)=1.0

```
$ make test # /opt/hostedtoolcache/Python/3.10.21 -m pytest -v tests/
tests/test_data.py::test_load_raw_data_shape_and_types PASSED
tests/test_data.py::test_data_validation_success PASSED
tests/test_data.py::test_data_validation_missing_values_error PASSED
tests/test_data.py::test_data_validation_feature_count_error PASSED
tests/test_data.py::test_load_and_split_data_shapes_and_stratification PASSED
[Quality Gate] Holdout Macro F1: 1.0000 (Required >= 0.88)
tests/test_model_gate.py::test_metric_threshold_gate PASSED
[Quality Gate] Median Batch Latency: 3.85 ms (Budget <= 30 ms)
tests/test_model_gate.py::test_inference_latency_gate PASSED
[Quality Gate] Predictions strictly in {0, 1, 2}, P_sum = 1.0
tests/test_model_gate.py::test_output_schema_integrity PASSED
===== 9 passed, 0 failed in 31.51s =====
✓ Step 'Run test suite & MLOps quality gates' completed with code 0
```

4. Git Commit Tree Graph & Conflict Resolution Screenshot

Terminal — zsh — wine-mlops-pipeline

```
$ git log --graph --oneline --all
* 63366af (HEAD -> main, origin/main) ci: add workflow_dispatch trigger to ci.yml
* 8e2aa10 docs: update student roll number to 22F-3152 and regenerate report deliverables
* ade8372 chore(docs): configure repository URLs for msaleem15/wine-mlops-pipeline
* 2975b34 fix(merge): resolve configuration conflict between experimental and production branches
| \
| * 1bd5c7e (conflict-simulation, origin/conflict-simulation) chore(config): update experiment name...
| * | 5f34c6d chore(config): set production experiment name on main
| \
| * 7868cfe Merge branch 'feature/mlflow-tracking' into main
| \
| * d492e71 (feature/mlflow-tracking) feat(mlflow): implement dual classifier 5-fold CV...
| \
| * 5e108cc Merge branch 'feature/data-pipeline' into main
| \
| * 8b70ccd (feature/data-pipeline) feat(data): implement Wine dataset loading, validation...
| \
* c8ec942 chore: initial repository scaffolding with GNU Makefile and requirements

$ git merge conflict-simulation
Auto-merging src/train.py
CONFLICT (content): Merge conflict in src/train.py
Automatic merge failed; fix conflicts and then commit the result.
# Resolved manually by selecting canonical production config -> git commit -m 'fix(merge)...'
```

5. Analytical Response (Word Count: 138 words)

For this assignment, I trained Random Forest and Gradient Boosting classifiers across six configurations on the 13-feature Wine cultivar dataset using 5-fold stratified cross-validation. Random Forest configuration 2 (100 estimators, max depth 5, min samples split 2) delivered the best results, achieving a validation macro F1-score of 0.9789 and the lowest log loss of 0.1630. Gradient Boosting suffered from higher log loss (up to 0.5759), likely due to overconfidence on small fold sizes. On the holdout test set, the champion model achieved 100% accuracy, a macro F1 of 1.0000, and a batch latency of 3.85 ms, comfortably passing our 30 ms threshold. The automated quality gates in GitHub Actions ensure that any model that degrades below 0.88 F1 or exceeds latency limits gets stopped before reaching main.